PrimusCredence

Provable Security for Applications

Not where your system was tested — what cannot happen in it.

1. **Access control**
   who may do what
2. **Network isolation**
   what can reach where
3. **Data residency**
   where data can go

primuscredence.com

UAE

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22+ YEARS OF EXPERIENCE IN PROVABLE SECURITY

PhD, IISc Bangalore
Model checking for information-flow security

Oracle Labs · 2014–2019
Java vulnerability detection

ConsenSys R&D · 2019–2021
Cross-chain protocols, consensus verification

Supra · VP of R&D · 2021–ongoing
BFT design & FV, bridges, DeFi & Crypto

TECHNIQUES

Static analysis, theorem proving, model checking

TOOLS

Dafny, TLA+, IVy, Lean, Isabelle

Where the Risk Actually Lives

The cloud base is proven. The app above it — and the seams between — are not.

Third-party plugins & extensions 1
Run inside the app — and inherit its privilege

APPLICATION 1

Its own roles, API tokens and public users — the authorisation logic and configuration in its database and admin panel

OWASP #1 Broken Access Control · #2 Misconfiguration — in 100% of apps

THE APP ↔ CLOUD & DATA SEAM

1 Access via APIs 2 Network paths & zones 3 Data paths & copies

CLOUD & INFRASTRUCTURE

VMs · containers · databases · queues · object storage · IAM · network

Access control & network — proven: Zelkova, Tiros

UNPROVEN TODAY

Where the breaches are

The app's own access control, its plugins, and the access, data and network seams beneath it.

→ Services 1 · 2 · 3 prove it.

PROVEN BASE

Cloud IAM & network — already formally verified.

OWASP 2025 — Access Control & Misconfiguration

The two categories at the top are found in 100% of apps tested.

A01

Broken Access Control

#1 — unchanged

40 CWEs · 1.84M occurrences · 32,654 CVEs

SSRF now folded in — easy to exploit, widespread.

A02

Security Misconfiguration

#2 — up from #5

16 CWEs · 719,084 occurrences

Again, 100% of tested apps carried a form of it.

BROKEN ACCESS CONTROL IN THE WILD

Capital One	106M	WAF SSRF → over-privileged role; a tipster found it, ~4 months in
Optus	9.5M	access-control bug on a dormant, internet-facing API — latent ~4 years
T-Mobile	37M	API data retrieval without authorization — ~6 weeks to notice
Dell	49M	auto-approved fake partner, then service-tag enumeration
Trello	15M	unauthenticated email lookup through rotating proxies
Twilio Authy	33M	unauthenticated endpoint enumerated for phone numbers

Each system did exactly what its access rules allowed — the rules were wrong.

Segmentation Is Asserted, Not Proven

Once inside, lateral movement turns an incident into a breach.

29 min
average eCrime breakout in 2025: initial access to lateral movement

27 sec
the fastest breakout ever observed

82%
of detections in 2025 were malware-free

Source: CrowdStrike 2026 Global Threat Report (2025 data)

WHY PROBING CAN'T ANSWER IT

- Tests only the paths you thought of
- Can't tell blocked from about-to-change
- East-west is claimed more than shown

REACHABILITY IS A CONFIG PROPERTY

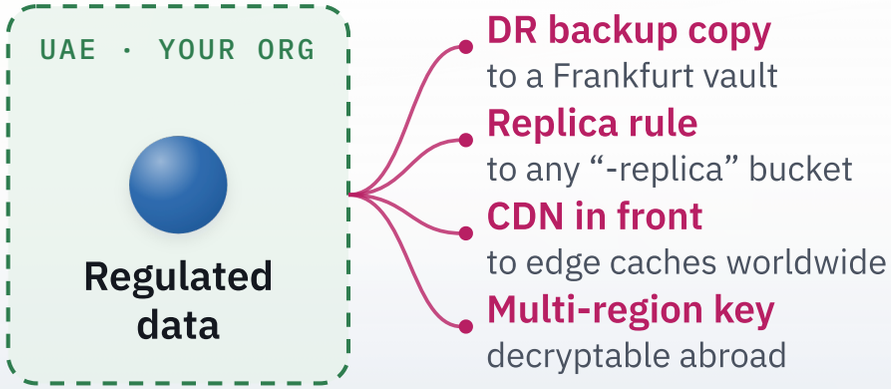
- Routes, SGs, NACLs, firewall rules
- Decided statically — no scanning window
- Precedent: AWS Tiros · Batfish

On OT networks active probing is often off-limits — the IT/OT boundary must be proven.

Residency Is Reviewed, Not Proven

Every change is reviewed alone — nobody re-checks where the data can now go.

FOUR QUIET WAYS OUT



WHY THE TOOLS STAY GREEN

- Region locks govern where, not whose
- DSPM shows where data is, not where it can go
- CSPM checks one resource at a time — and rewards cross-region backups

€1.2B

Meta, 2023: EU users' data sent to the US (Irish DPC)

€530M

TikTok, 2025: EEA users' data remotely accessed from China (Irish DPC)

€290M

Uber, 2024: EU drivers' data sent to the US (Dutch DPA)

A region lock answers where. Sovereignty also asks whose.

AI Industrialises the Attacker

Point-in-time testing cannot keep pace with an enumerating attack.

54M
lines of code analysed by autonomous AI systems (DARPA AIxCC 2025)

$3.52
compute cost per LLM-agent exploit attempt on a real CVE (Fang et al. 2024)

87%
of 15 real CVEs exploited from the public advisory alone (Fang et al. 2024)

TRADITIONAL PEN TEST

10^2–10^3 paths, human-selected
2–4 week window, then closed
High marginal cost per path
“We found no way in”

AI-DRIVEN ADVERSARY

10^5–10^7 paths, machine-enumerated
Continuous, no window
Near-zero marginal cost
Approaching exhaustive

Path counts are orders of magnitude, for illustration.

Proof is the one capability that answers exhaustive search in kind.

What Today's Security Stack Leaves Open

Each tool answers its own question well. One question stays open.

SAST	Knows code patterns, not intent. A check present but wrong still passes.
DAST	Tests the endpoints and personas it is given — not every user × object.
IAST / RASP	Same-code-path BOLA is invisible — was the call even permitted?
DSPM	Finds where regulated data is — never where it can go.
CNAPP / CSPM	Stops at the cloud IAM role; never reads the app's own roles.
WAF / API sec	Judges traffic, not grants: whether it should be permitted is upstream.
Pen test	Shows a way in, or that none was found — not that none exists.
SOC · SIEM	Sees what happens — not what the rules would allow next.

They all share one gap: no statement of what is allowed, decided over every request.
Keep every one of them — and add the layer that closes that gap.

Proof — Now Feasible

Formal verification used to take expert teams years. AI brings it into range.

THEN

- Expert teams, bespoke specs, years
- Confined to avionics, silicon, protocols
- “End users won’t write a spec”

NOW — AI-ASSISTED

- AI assists modelling & proof search
- Expert owns & ratifies the spec
- AWS: ~1B SMT queries a day (CAV 2022)

What we add A preventive control with evidence behind it, beside your testing and SOC.
We add to your stack; we don’t replace it. Proof alone isn’t enough.

WHAT STAYS WITH YOU — ALL OF IT NEEDED

Configuration drift, shadow assets and anything outside the model

Insider misuse, and valid credentials used exactly as intended

Detection, response and the rest of the programme you already run

Automated Reasoning

Proof establishes that bad things cannot happen.

1. **Formalise**
   Your existing artefact becomes a logical model
2. **State**
   The bad state you fear becomes a formula
3. **Solve**
   Z3, CVC5 and Batfish decide every case
4. **Report**
   A proof, or a reproducible witness

AI ASSISTS EVERY STEP · THE EXPERT OWNS THE SPEC

UNSAT

No violating state exists — over the entire space, under a stated model.
Not “none was found”.

SAT + witness

The exact request, path or chain that breaks it — reproducible on your estate.

Not every property is decidable, and a proof holds only for the property and model stated.
Industrial practice: AWS · Microsoft · Meta · Airbus.

Services

One method — three services: access control, network isolation, data residency.

1 ACCESS CONTROL

App entitlement proofs

Every app role, token, plugin route and public user — decided over all requests.

Who may do what

2 NETWORK ISOLATION

Network segmentation

Routes, security groups, firewall and Kubernetes rules — every path between zones.

What traffic can reach where

3 DATA RESIDENCY

Residency proofs

Backups, replicas, keys, caches — not where data is, but where it can go.

Where regulated data can go

EVERY SERVICE ENDS in a proof — or the exact request, path or chain that breaks it.

1. Application Access-Policy & Entitlement Verification

> *Can any app role, API token, plugin route or public user reach data it must not — through the app or its cloud role?*

THE PROBLEM

Authorisation lives in the app's admin panel, its plugins and its one cloud role. Each layer passes its own review. Nobody checks them together.

Checked at the moment an admin clicks Save — before anyone can use the change.

ENGINE	Cedar + SymCC (CVC5), from the live app
PRECEDENT	AWS Cedar Analysis
BUYERS	Platform & AppSec teams · ISVs

IN THE DEMO

tools run first; none decides these

VIOLATED **Two admin changes, each fine**
Together, an Auditor at one hospital exports another hospital's restricted records. HTTP 200.

VIOLATED **The GraphQL plugin side door**
No login, any document. REST still says 403.

VIOLATED **The app ↔ cloud seam**
One cloud role for all users: files cross tenants.

PROVED **after each fix — every user, every request**

OUTPUT A proof that no role, token, plugin route or public user reaches what it must not — or the exact request that breaks it.

SERVICE 2 — WHAT TRAFFIC CAN REACH WHERE

2. Network Reachability & Segmentation Proofs

> *Does any path exist from an untrusted zone to this one — and does east-west segmentation actually hold?*

THE PROBLEM

Probing tests the paths you thought of. Reachability is a property of the configuration — routes, firewall rules, NAT — across every segment at once.

No scanning window, no production traffic — safe on OT, and re-run on every change.

ENGINE	Batfish + Z3, from config files
PRECEDENT	AWS Tiros → VPC Reachability Analyzer
BUYERS	Finance · OT & CNI · sovereign cloud

IN THE DEMO

tools run first; none decides these

VIOLATED **One policy, every zone pair**

FortiGate/OT and AWS. AWS → historian → SCADA, hop by hop, with the rule and its owner.

WOULD VIOLATE **A change, before the CAB**

A pending firewall rule opens 2 new paths into OT.

VIOLATED **One dash in a NetworkPolicy (optional)**

Any namespace reaches tenant B's database on 5432.

PROVED **every row, after the fix — no packet sent**

OUTPUT A proof that no path exists — or the path, hop by hop, with the rule that permits it and the owner of that rule.

SERVICE 3 — WHERE REGULATED DATA CAN GO

3. Data Residency & Sovereignty Proofs

> *Can regulated data leave the country or the organisation — or could anyone make it leave?*

THE PROBLEM

Residency is asserted by configuration review. Backups, replicas, caches and keys are each reviewed alone. DSPM shows where data is, not where it can go.

Not where your data is — where it can go.
Today, and whatever anyone could set up.

ENGINE	SMT over IAM, SCPs and key policies
PRECEDENT	AWS Zelkova → IAM Access Analyzer
BUYERS	Sovereign cloud · government · banks

IN THE DEMO

tools run first; none decides these

AT RISK (can) **Nothing has left — yet**
An ops role can replicate the data to a "-replica" bucket outside the organisation. Replayed live.

VIOLATED (is) **A DR copy and a CDN**
A backup in Frankfurt; edge caches outside the UAE.

OUTSIDE MODEL **Application egress**
Shown, not hidden: the app can reach the internet.

PROVED **for storage, keys and edge after the fix**

OUTPUT A proof that no channel moves the data outside the approved jurisdiction or organisation — or the chain: store, mechanism, role, policy line, destination.

Our Approach

We partner with cybersecurity solutions providers, adding provable assurance.

HOW WE PARTNER — THROUGH THE PROVIDERS

Provable control effectiveness inside their bids.

They keep the client relationship and the contract.

We don't certify — a channel, not a rival.

WHOM WE REACH THROUGH THEM — THE GCC CLIENT BASE

Government & sovereign	Federal & emirate entities · Digital-government programmes
Aviation & CNI	Air navigation · Airports · Utilities
Energy, oil & gas	National energy companies · OT operators
Banking & finance	Regional banks · Sovereign funds · FinTechs
Telecom & cloud	Telecom operators · Sovereign-cloud providers

Why the Provider's Clients Will Ask

Four things happened at once in the GCC — and their clients feel all four.

Mandatory replaced voluntary

UAE cyber-resilience obligations now carry penalties of AED 100k–3m.

Assurance shifted to evidence

Regulators increasingly ask for demonstrable control effectiveness, not attestation.

Third-party risk is now in scope

Rules increasingly require evidence that integrations and components are controlled.

Sovereignty must be demonstrable

Localisation asks whether regulated data can leave — our residency proofs settle it.

WHERE A FINDING ALSO LANDS

DESC ISR	1 2 3	NESA / UAE IAS	2 3	UAE PDPL	2 3
SAMA CSF	1 2 3	NCA ECC	1 2 3	PCI DSS 4.0	1 2
ISO 27001	1 2 3	CIS Controls	1 2 3	NIST CSF 2.0	1 2 3

Regulation routes the finding to a second reader. It is not the reason to commission it.

What a Provider Can Resell

We do not certify — so an audit firm or an MSSP is a channel, not a rival.

Detection · response · posture reporting **+** **Provable control effectiveness**

THE COMMERCIAL MOVE

- Proof inside bids you already win
- Recurring: re-proven on every change
- You keep the client and the contract

WHY IT DEPLOYS EASILY

- Read-only access, no agent on any host
- No scanning window, no production traffic
- Verdicts can land in your SIEM and tickets

We produce the technical evidence. Your assessor keeps the opinion.

One Result, Three Registers

One analysis, read by engineering, risk and audit — each in its own terms.

First line

Engineering and security operations

The property, its model and assumptions, and either the proof or a reproducible attack path.

Second line

Risk

Likelihood moved off a qualitative guess, with residual risk and the model's limits stated.

Third line

Internal audit and compliance

A clause-mapped control assertion: objective, status, owner and re-verification date.

✓ Design → operating effectiveness · attaches to your GRC records · assessors probe less

The first register is the product. The second and third make it worth more than it cost.

Take Away

Bringing mathematical rigour to security —
proof alongside testing, from the app to the cloud.

The shift

Attack path discovery is automated, parallel and cheap. Detection shows what is happening; proof shows what cannot happen.

The addition

We do not replace testing, monitoring or the SOC. We make one class of controls machine checkable.

The honesty

A proof is relative to a model and a specification — both stated. We produce the evidence; we do not certify.

START HERE — MICRO ASSESSMENT OF SERVICE 1

One narrow, high-risk slice. Read-only access, no agent deployed, weeks not months — and a counter-example on your own systems, or a proof.

Thank You

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